Database Application and Design (SOC-3060)

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WEEK 11

LDKR
Language, Data, Knowledge, Reasoning

DATABASE APPL. AND DESIGN

Logical Data Design and SQL

WEEK 11

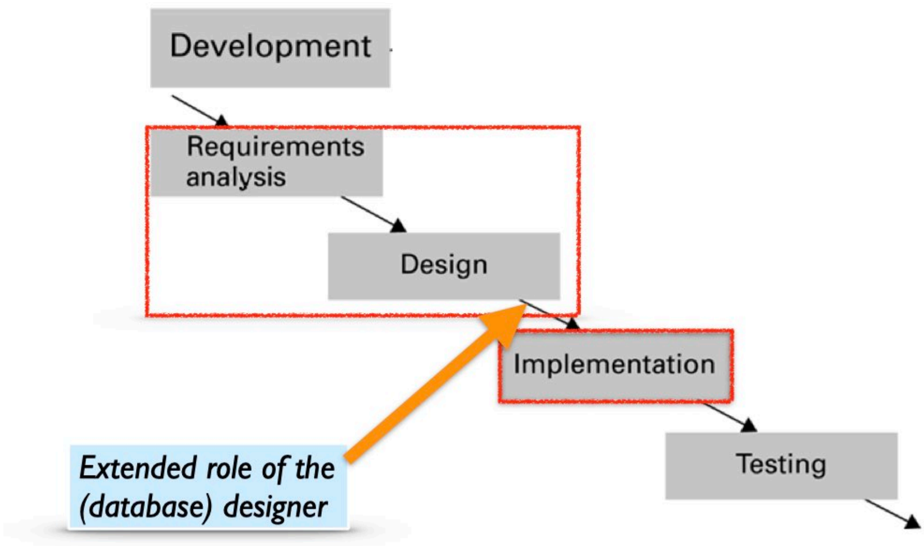
Code: W11.L1L2[LAB#7]

Outline

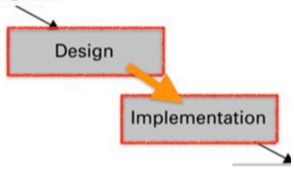
Any question from you?

...

Designer in Software Life Cycle



From Data Model to Implementation



- Two phases (pre-implementation):
  - **Logical-design (phase II)** : the designer refines/completes the logical schema w.r.t the database system (DBMS) that will be used.
  - **Physical-design** : the designer uses the resulting **system-specific database schema** from logical-design to specify the physical features of the database. These features include the form of file organization (indices), storage structures, etc.

DB Design: Main Steps

- Heart of the database design (development) process:

  1. **Requirements collection and analysis (Phase 1)**
  2. **Conceptual database design (Phase 2)**
  3. **Choice of DBMS**
  4. **Logical design (or: data model mapping) (Phase 4)**
  5. **Physical design (Phase 5)**
  6. **System implementation and tuning ("re-design")**

DBMS-independent

DBMS-dependent

Phase 4: Logical Design

- Also called **data-model mapping**.
- Goal is to create a logical schema and external schemas (views) in data model of selected DBMS…

  … for us: DBMS based on the Relational Model (Codd).

- The mapping consists of two stages:
  1. System-independent (*optimized*) mapping:
     – ER /EER diagrams mapping into the chosen model' schemas
  2. Tailoring schemas to a specific DBMS
- Deliverable: **DDL statements in chosen DBMS language**

Outline

Alternative Notation for Relationships

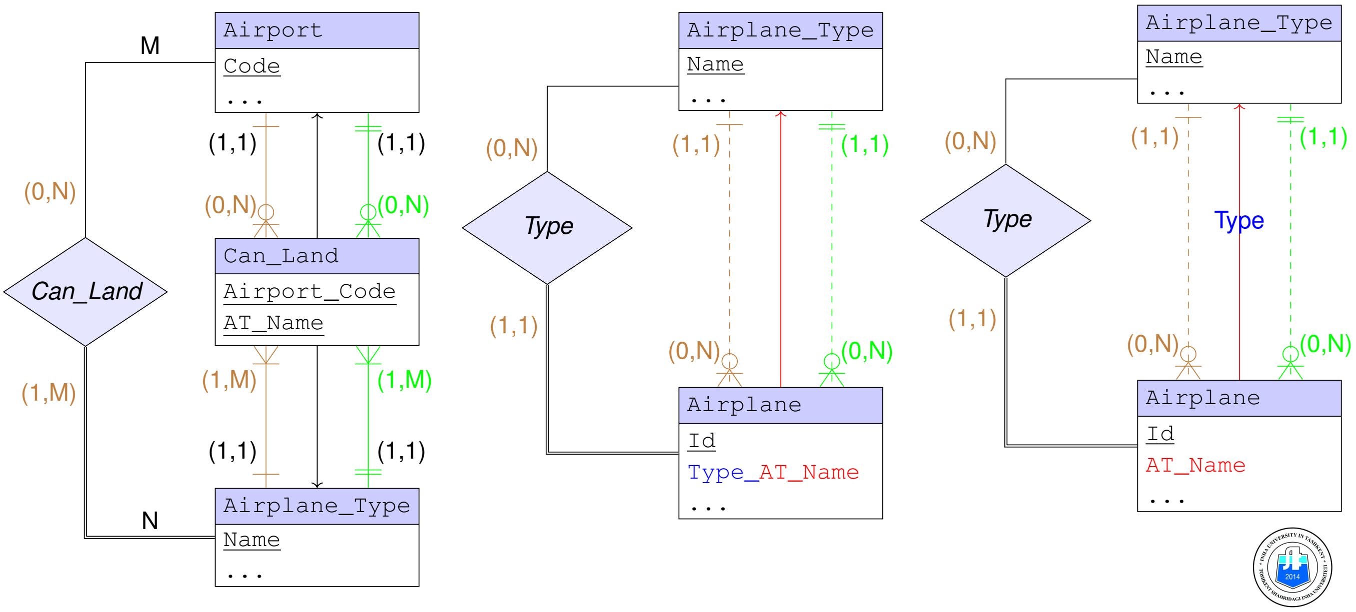
Used within CASE Tools

CASE: Computer-Aided Software Engineering

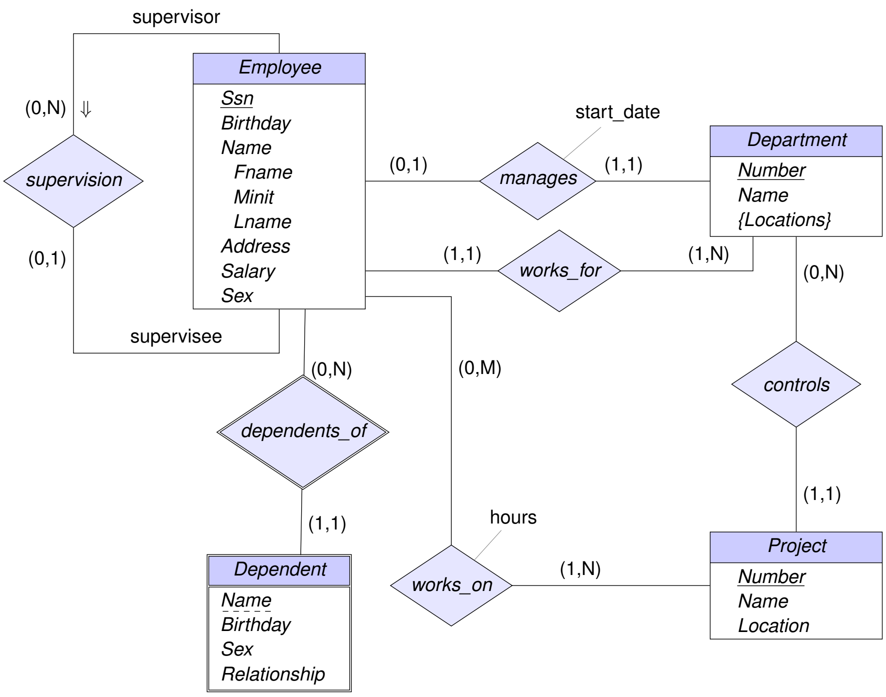
ER-Win’s and MySQLWB’s Notation for (*min*, *max*) Constraints

M:N Relationships:

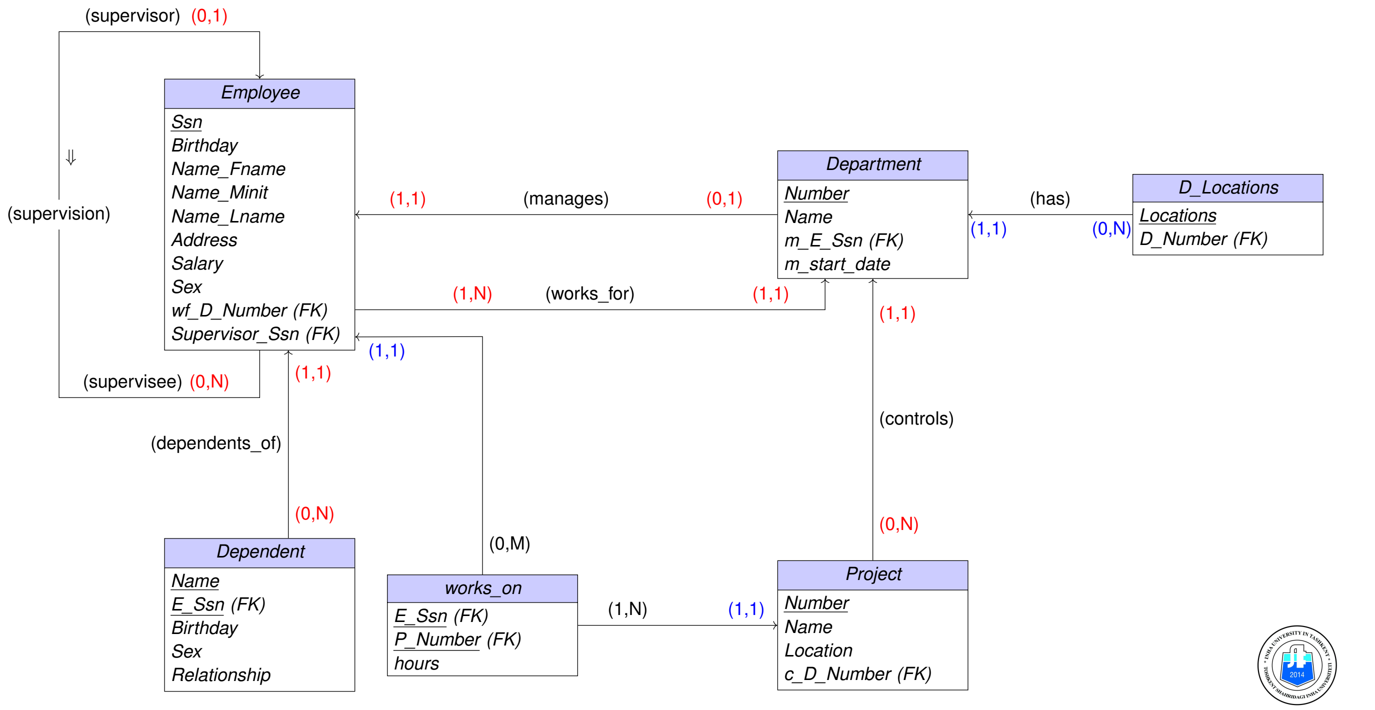
1:N Relationships:



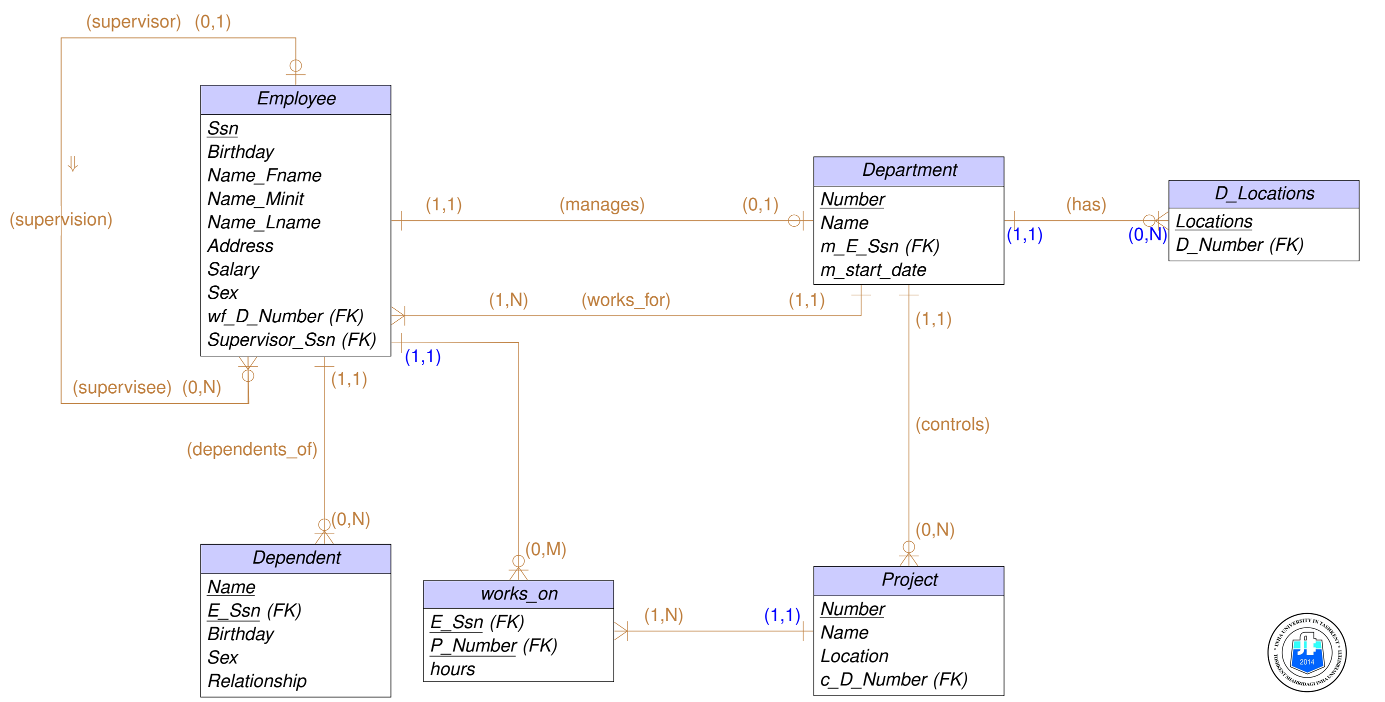
Example: E-R Diagram for Company DB



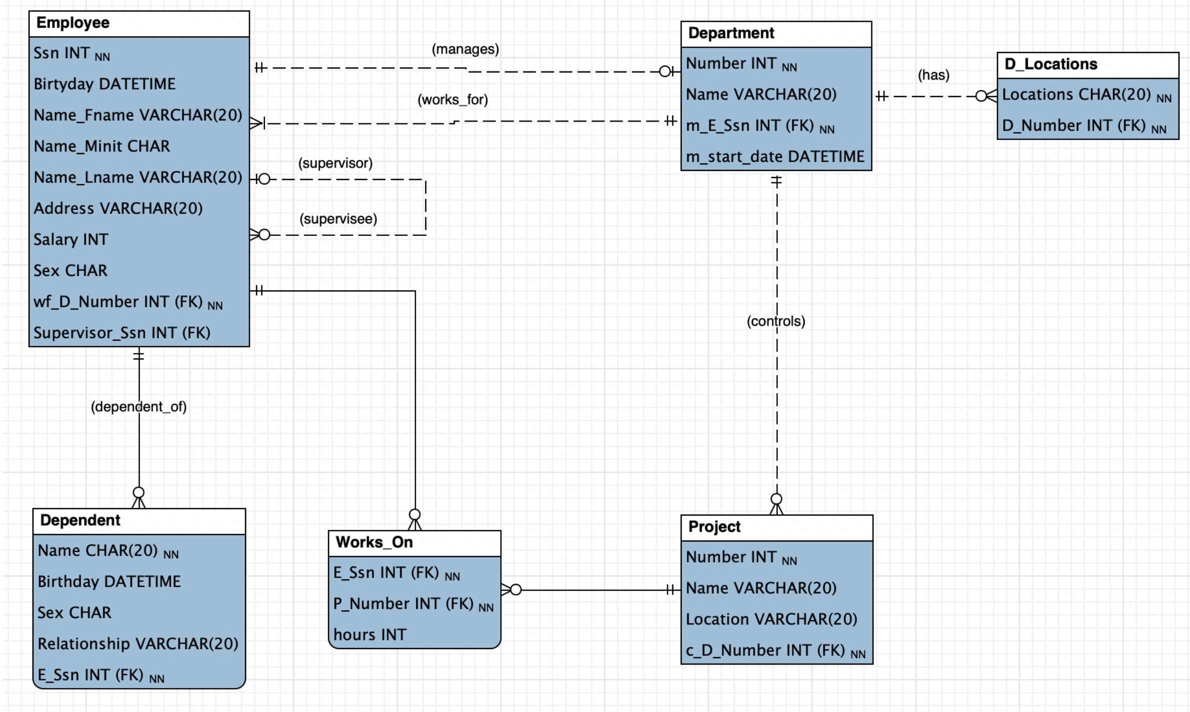
Optimized Logical Schema (tagged with (*min*, *max*) contraints)



Example (*cont.*): “Crow’s Foot” Notation

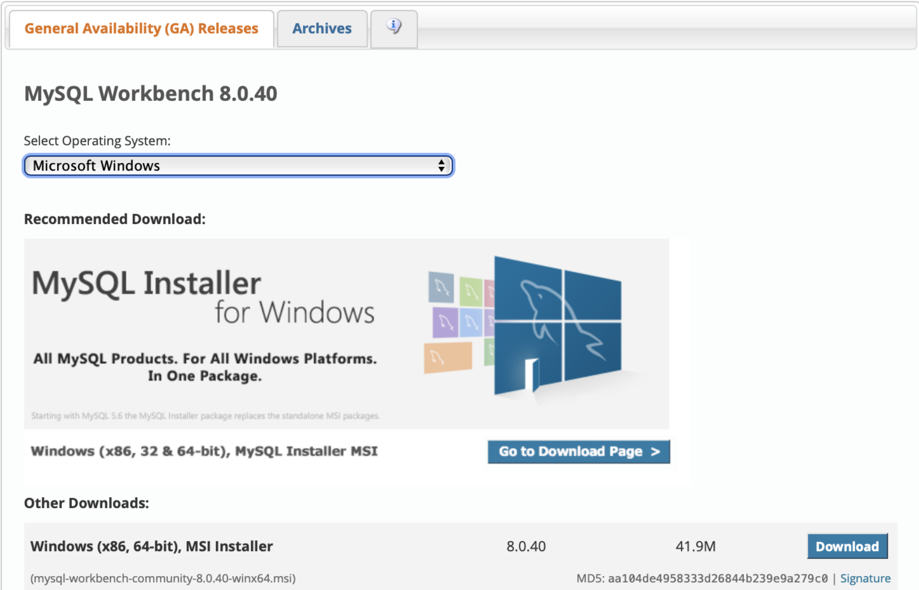


Example (*cont.*): Solution by MySQL Workbench



Outline

Source: https://dev.mysql.com/downloads/workbench/



Remark: At this point we assume that students and teams downloaded MYSQL WORKBENCH and installed it to their laptops. You need it to code the logical design by SQL.

WARNING!

The kind of diagrams for conceptual design that we have studied so far are generally richer (i.e., more expressive and complex) than the diagrams that can be expressed in CASE tools like *ER-Win* or *MySQL Workbench.*

As a consequence, each element of conceptual diagrams (EE-R model) that can't be directly expressed. It means that **aggregation**, **union class**, and **IS-A relationships** must first be mapped into the relational model, in particular by using the "crow's foot" notation.

IMPORTANT: *On the limits of CASE tools*

- "Multivalued" and "composite" attributes in E-R/EE-R diagrams ***must*** first be transformed according to logical design.
- "1-N relationship sets with attributes" in E-R/EE-R diagrams ***must*** first be transformed (and possibly optimized) according to logical design (data model mapping).
- "N-ary relationship set" in E-R/EE-R diagrams ***must*** first be transformed into the equivalent set of binary relationships.
- "Many-to-many relationship sets with attributes ***must*** first be transformed according to logical design (data model mapping).

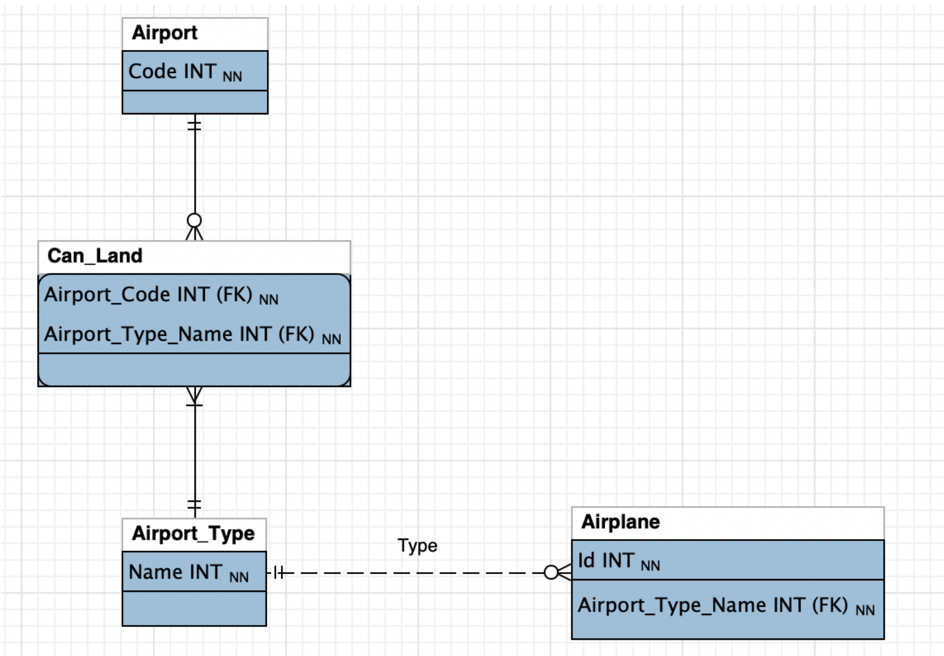
On the limits of CASE tools (cont.)

- "Aggregate entity sets" (aggregations) in EE-R diagrams ***must*** first be transformed according to logical design.
- "Subclasses" in EE-R diagrams ***must*** be transformed according to logical design whatever is the mapping's method (M1…M4).
- "Union classes" (categories) in EE-R diagrams ***must*** first be transformed according to logical design (data model mapping).

Example

Conceptual Design by Reverse Engineering

Example



Practice in class with Instructor - no slides.

End of Lecture 1

Thank you!

Outline

LAB #7: Logical design by SQL*

LECTURE 2

* Participation checked (2-3 members per team only allowed)

In class only (no slides provided).

End of Presentation

Thank you!

Questions?

Summary

NB: In red are topics on **Video Lecture with mandatory attendance** (eClass).

DATABASE APPL. AND DESIGN

End of Slides

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